

UCS671: Edge AI and Robotics: Embedded Vision

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Course Objective: This course will provide students with advanced conceptual knowledge and practicals on various computer vision and deep learning applications and provide the overall environment for end-to-end pipeline development from data collection to deployment.

Syllabus

Introduction: Utilizing Jetpack SDK and other NVIDIA Toolkits to deploy CNN models on Jetson, Creating Jetbot kits and deploying various applications, Working with NVIDIA Robotics toolkit: Isaac SIM SDK and Gazebo for collision avoidance, path following.

Introduction to Edge AI: AI at the Edge & IoT, Jetson Architecture, Getting Started with Jetpack, NGC Containers in Jetson, Getting started with NGC & Containers on Jetson.

Introduction to NVIDIA Toolkits and SDKs: Transfer Learning Toolkit, TensorRT Optimization, Triton Inference Server for Inferencing and Deployments, Various Visualization Tools, Kubernetes Deployment, Deepstream SDK, Deploying Classification, Detection and Segmentation CNN Models on Jetson Devices.

Perception & Autonomous Navigation: Building JetBot Kits, Introduction to basic motion on JetBot, Collision Avoidance: Stop/Go classifier (JetBot), freespace detection, Path Following: Recording user input/video + DNN training (DriveNet), Simulation: Gazebo & Isaac SIM.

Advanced Vision & SLAM: Pose Recognition (Deploying Human pose model), Depth Estimation: Mono/Stereo depth and point extraction, Visual Odometry: Camera pose estimation from DNNs, SLAM on JetBot.

Laboratory Work:

- Setting up the Jetson Project kit.
- {DLI Online Course: Getting Started with AI on Jetson Nano.}
- Deployment of Various Classification, Object Detection and Segmentation models using TensorRT, and Triton Inference Server in Jetson Nano.
- Getting started building various Jetbot Kits.
- Basic Motion with Jetbot
- Collision Avoidance with Jetbot kit
- Object following and Road following (DriveNet) with Jetbot.
- Teleoperation with Jetbot.
- Human Pose Estimation in Jetson Nano/JetBot.
- Implementing SLAM on Jetbot.

Course Learning Objectives (CLO)

After the completion of the course the student will be able to:

1. Introducing concepts of Edge computing and Jetson kits.
2. Provide understanding of approaches, concepts and algorithms used in computer vision deployments.
3. Practice implementation of various concepts and tools covered in the course.

- 4.Utilize programming and scientific tools for relevant software implementation.
- 5.Understand GPU computing for building advanced computer vision pipelines on Jetson devices.

Text Books

1. Computer Vision: Algorithms and Applications, R. Szeliski, Springer, 2011.
- 2.Computer Vision: A Modern Approach, D. Forsyth and J. Ponce, Prentice Hall, 2nd ed., 2011.
- 3.Richard Szeliski, Computer Vision: Algorithms and Applications, Springer-Verlag London Limited 2011

Reference Books:

1. Richard Hartley and Andrew Zisserman, Multiple View Geometry in Computer Vision, Second Edition, Cambridge University Press, March 2004.
- 2.K. Fukunaga; Introduction to Statistical Pattern Recognition, Second Edition, Academic Press, Morgan Kaufmann, 1990.
- 3.R.C. Gonzalez and R.E. Woods, Digital Image Processing, Addison- Wesley, 1992.
- 4.Christopher M. Bishop; Pattern Recognition and Machine Learning, Springer, 2006.